

AAYAN YADAV

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🎓 [Google Scholar](#)

🌐 [linkedin.com/in/aayanyadav09](https://www.linkedin.com/in/aayanyadav09)

🐙 github.com/ydvaaayan

🌐 [Portfolio](#)

Education

*GPA conversion to 4.0 scale done according to this website: <https://www.scholaro.com/gpa-calculator/>

Indian Institute of Technology, Roorkee

Oct. 2022 – May 2026

Bachelor of Technology in Data Science and Artificial Intelligence

Uttarakhand, India

CGPA : 9.03/10.0 (~ 3.826/4.0)*

Experience

AuraML | Research Intern (Hybrid)

April 2025 - August 2025

- Developed a scalable pipeline combining VLMs and graph diffusion models to generate and edit large 3D warehouse scenes from text prompts with efficiency and accuracy.

Proteus Photos | Research Intern (Remote)

July 2024 - February 2025

- Built a classifier for detecting AI generated images. Improved upon SoTA classifier by finetuning contrastive language-image based encoder. Also experimented with some few shot techniques and adversarial training.

University of Michigan | Research Collaboration (Remote)

September 2023 - March 2024

- Worked on refining instance segmentation masks of MS-COCO dataset and analysed its impact by training and benchmarking models. I also wrote and presented a full paper accepted at ECCV 2024 with my amazing coauthors.

Publications

* denotes equal contribution.

Benchmarking Object Detectors with COCO: A New Path Forward [\[Website\]](#)[\[Preprint\]](#)[\[GitHub\]](#)

Shweta Singh*, Aayan Yadav*, Jitesh Jain, Humphrey Shi, Justin Johnson, Karan Desai

ECCV 2024

- Created refined annotations for the COCO-2017 dataset to mitigate imperfections by leveraging the Segment Anything model and LVIS dataset in a semi-automatic pipeline. Benchmarked object detectors to validate annotations.

StegaVision: Enhancing Steganography with Attention Mechanism (Student Abstract) [\[Preprint\]](#)[\[GitHub\]](#)

Abhinav Kumar, Pratham Singla, Aayan Yadav

AAAI 2025

- Explored the impact of various attention mechanism i.e. channel and spatial and their permutations on encoder-decoder architecture for steganography.

Provenance Detection for AI-Generated Images: Combining Perceptual Hashing, Homomorphic Encryption, and AI Detection Models [\[Preprint\]](#)[\[GitHub\]](#)

arxiv 2025

Shree Singhi*, Aayan Yadav*, Aayush Gupta, Shahriar Ebrahimi, Parisa Hassanizadeh

- A content provenance system to detect AI generated images. I worked on training adversarially robust binary classifier for the task. I leveraged embedding space of large self supervised vision-language models.

Projects

SLAFCoM: A Study on Loss Functions for Adversarial Finetuning of Contrastive Models [\[GitHub\]](#) April 2025

- Introduced a Clean Consistency Term in the loss function and experimented to improve adversarial finetuning.

Sirius | Pathway, OpenAI Swarm [\[GitHub\]](#)

November 2024

- Build an agentic RAG system using SoTA techniques like AdaRAG, PlanRAG, HyDE, SPLADE, MetRAG, RRF etc.

Achievements

Sesquicentennial (1997) Batch scholarship for Academic Excellence Awarded to one out of 1.3k+ students in IIT Roorkee for demonstrating academic excellence and research aptitude.

All India Rank 1353, JEE Advanced 2022 150k+ students in India take this for admission in IITs.

All India Rank 1962, JEE Main 2022 1M+ students in India take this for JEE Advanced eligibility.

All India Rank 993, KVPY(SX) 2022 Prestigious national fellowship for students with high research aptitude.

NTSE Scholarship Awardee, 2021 NTSE is a national-level examination for the secondary school level to identify and recognize students with high intellect and skills.

Extracurricular

Vision and Language Group, IIT Roorkee | Co-President

March 2023 – Present

Deep learning group on campus that organizes open discussions and projects related to deep learning.[\[Website\]](#)

Data Science Group, IIT Roorkee | Joint Seceretary

April 2023 – Present

Data Science and Machine Learning Group on campus that organizes workshops, talks and projects.[\[Website\]](#)

References

Prof. Justin Johnson | Email: justincj@umich.edu

United States

Assistant Professor at University of Michigan | Co-founder at World Labs

Dr Karan Desai | Email: kdesai@umich.edu

United States

Ph.D. in CSE at Universtiy of Michigan | Member of Technical Staff at World Labs

Prof. Sanjeev Kumar | Email: sanjeev.kumar@ma.iitr.ac.in

India

HOD of Mehta Family School of Data Science and Artificial Intelligence, IIT Roorkee

Arjun Gupta | Email: arjuntheprogrammer@gmail.com

India

Solution Architect at OpenAI | Former Co Founder & CTO at AuraML